CLOUDSECURITY

UNIT 3ASSIGNMENT

Set Up a Web Application on AmazonEC2

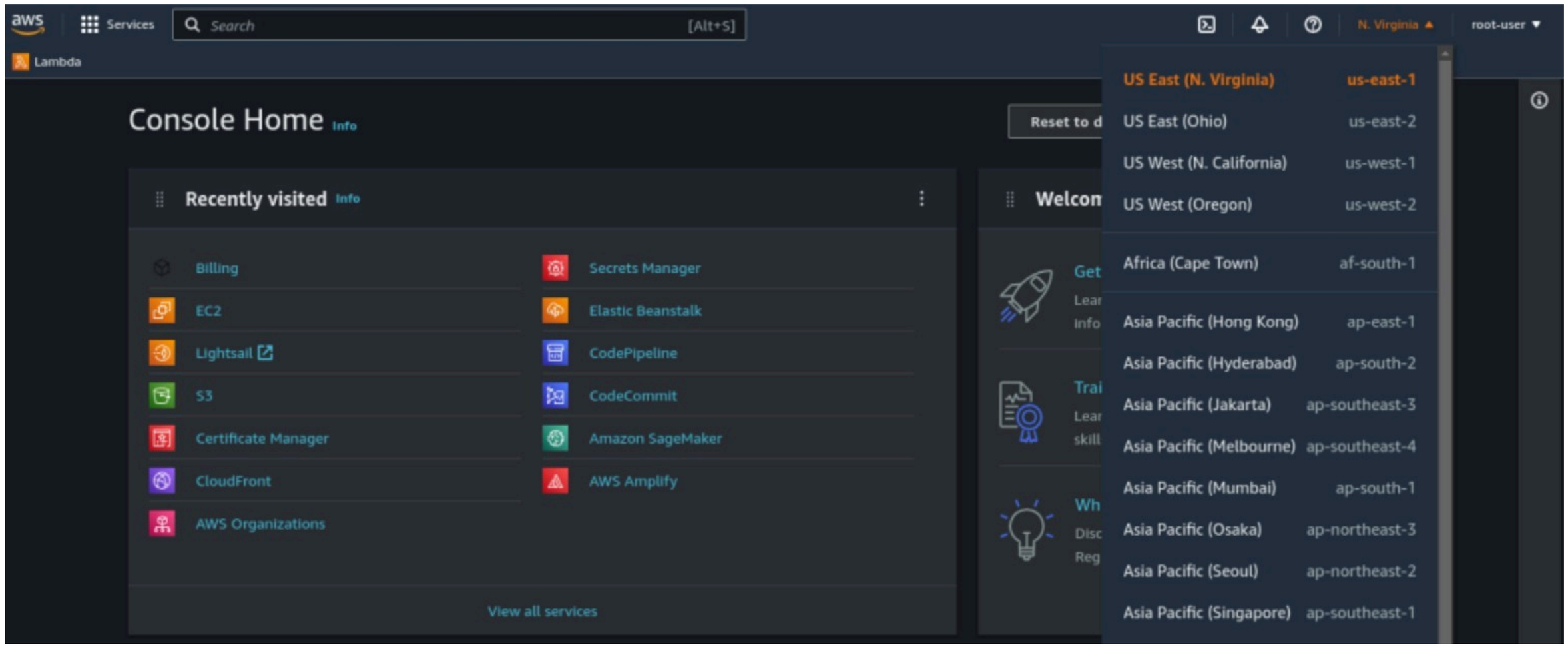
Step-by-steppracticalreportwithconfiguration,commands,expectedoutput,verification, and security considerations

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Subject	
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Execution and outputnote
Thisreportcontainscommandsandrepresentativeexpectedoutputsfordocumentation.Theoutputexamplesare illustrative and may differ by operating-system image, package version, region, or account settings. They are not represented as captured execution evidence.

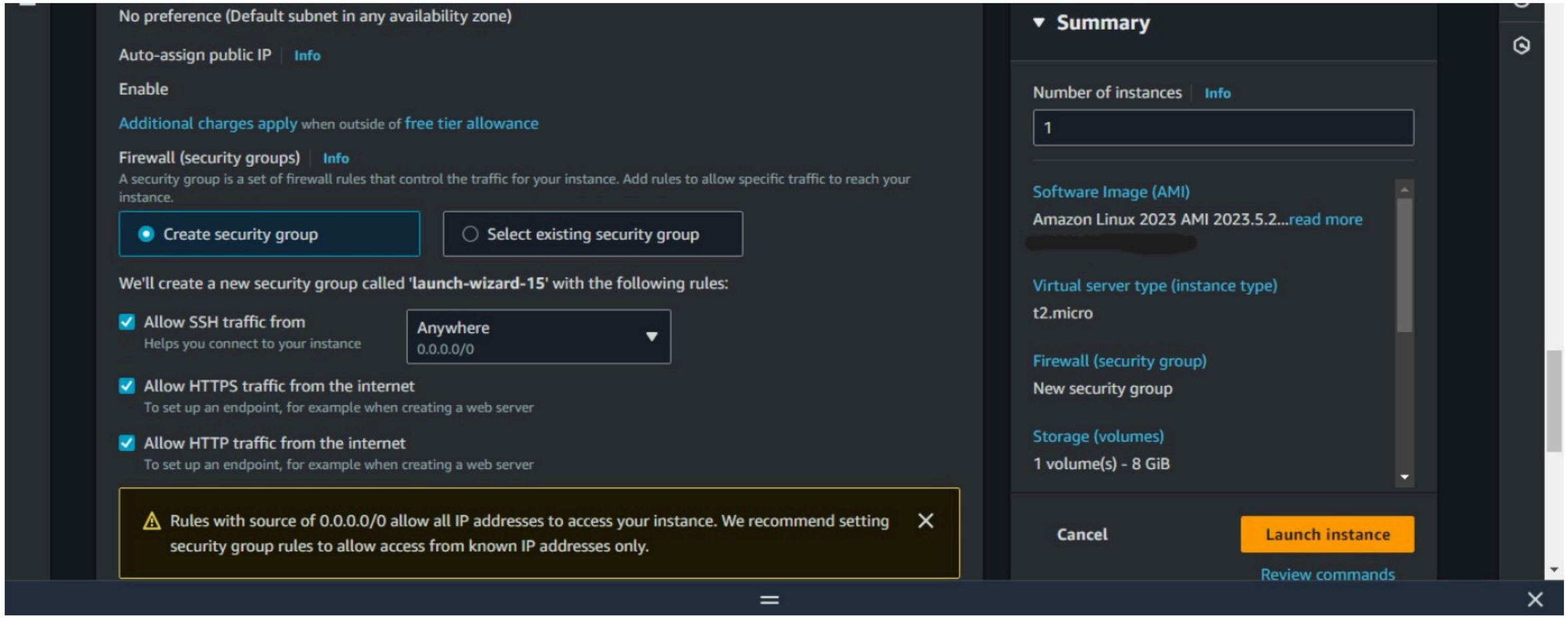
AWS Console regionselection

Screenshotevidence



EC2 launch instanceconfiguration

Screenshotevidence



Additional Terraform infrastructure configurationreference


```
variable "application_name" {}
resource "aws_launch_configuration" "application-launcher" {
  name_prefix = "${var.env}-${var.application_name}-lc-"
  image_id = "${var.instance_ami}"
  instance_type = "${var.instance_type}"
  ...
}
resource "aws_iam_instance_profile" "application_instance_profile" {
  name = "${var.env}-${var.application_name}-instance-profile"
  ...
}
resource "aws_security_group" "application_sg" {
  name        = "${var.env}-${var.application_name}-application-sg"
  ingress {
    from_port       = 22
  ...
}
resource "aws_autoscaling_group" "application_asg" {
  name = "${var.env}-${var.application_name}-asg-${aws_launch_configuration.application-launcher.name}"
  launch_configuration = "${aws_launch_configuration.application-launcher.name}"
  min_size = "${var.instance_min_count}"
  ...
}
```

1. Aim

TolaunchanAmazonEC2Linuxinstance,configurerestrictedSSHandHTTPaccess,installandstartthe Apache web server, deploy a simple HTML page, and verify that it is served over HTTP.

2. Requirements

- AnAWSaccountwithpermissiontocreateEC2instances,keypairs,andsecuritygroups.
- AnAmazonLinux2023EC2instanceinasuitableAWSRegion.
- AnSSHclient(Linux/macOSterminal,WindowsTerminal,oranotherSSHclient).
- Alocalprivatekeyfile(.pem)associatedwiththeinstance;keepitprivate.

3. Security GroupConfiguration

Inboundrule	Protocol/port
SSH	TCP22—source:MyIP(yourcurrentpublicIPv4address/32)
HTTP	TCP80—source:0.0.0.0/0forapublicdemonstrationwebsite;restrict further if required
Outbound	Keepthedefaultoutboundrulesunlessthelabrequiresastricterpolicy

Securitynote: NeversetSSHsourceto0.0.0.0/0foraroutinestudentlab.HTTPispublicinthis demonstration because the webpage must be reachable from a browser.

4. Procedure

Step 1 — Launch the EC2instance

- OpenAWSManagementConsole→EC2→Instances→Launchinstances.
- Choose Amazon Linux 2023, select an eligible instance type (for example, a small free-tier-eligible type if available to your account), and select or create an SSH key pair.
- Create or select a security group using the inbound rules above. Enable a public IPv4 address if you need to access the website from the internet.
- Launchtheinstanceandwaituntiltheinstancestateis Runningandstatuschecks pass.

Step 2 — Connect usingSSH

ReplacetheplaceholderswithyourrealkeypathandEC2publicDNSnameorIP.OnLinux/macOS,set restrictive key-file permissions before connecting.

```
chmod400my-key.pem
ssh-imy-key.pemec2-user@<EC2-PUBLIC-DNS-OR-IP>
```

Representative connection output (exact wordingvaries):

```
The authenticity of host'ec2-...compute.amazonaws.com'can't beestablished.
Areyousureyouwanttocontinueconnecting(yes/no/[fingerprint])?yes
[ec2-user@ip-10-0-1-25 ~]$
```

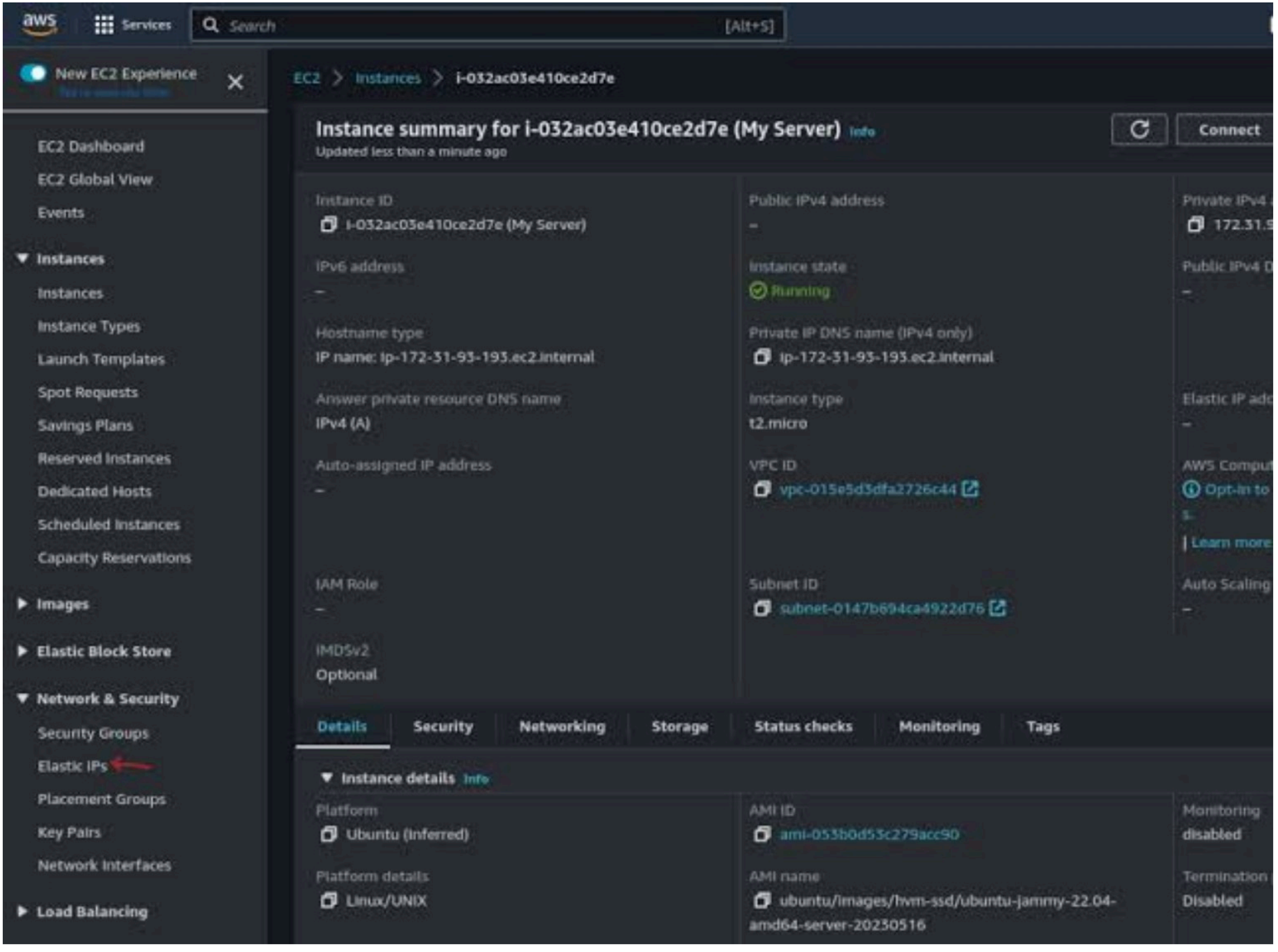
Step 3 — Install Apache and start theservice

```
sudodnfupdate-y
sudodnfinstall-yhttpd
sudosystemctlenable--nowhttpd
sudosystemctlstatushttpd--no-pager
```

Representative statusoutput:

```
httpd.service - The Apache HTTP Server
 Loaded:loaded(.../httpd.service;enabled)
```

EC2 instance summary and instancestate



Apache HTTP Server service status(apache2)


```
aaronkilik@tecmint ~ $ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese
   Active: active (running) since Mon 2017-09-04 10:05:51 EAT; 4h 8min ago
  Process: 3030 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=
  Process: 1182 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU
 Main PID: 1395 (apache2)
    Tasks: 7 (limit: 512)
   CGroup: /system.slice/apache2.service
           ├─1395 /usr/sbin/apache2 -k start
           ├─3037 /usr/sbin/apache2 -k start
           ├─3038 /usr/sbin/apache2 -k start
           ├─3039 /usr/sbin/apache2 -k start
           ├─3040 /usr/sbin/apache2 -k start
           ├─3041 /usr/sbin/apache2 -k start
           └─4232 /usr/sbin/apache2 -k start

Sep 04 10:05:39 tecmint systemd[1]: Starting The Apache HTTP Server...
Sep 04 10:05:50 tecmint apachectl[1182]: AH00558: apache2: Could not reliably
Sep 04 10:05:51 tecmint systemd[1]: Started The Apache HTTP Server.
Sep 04 10:10:43 tecmint systemd[1]: Reloading The Apache HTTP Server.
Sep 04 10:10:43 tecmint apachectl[3030]: AH00558: apache2: Could not reliably
Sep 04 10:10:43 tecmint systemd[1]: Reloaded The Apache HTTP Server.
lines 1-22/22 (END)
```

```
     Active: active(running)
```

Step 4 — Create a simple HTML webpage

```
sudotee/var/www/html/index.html>/dev/null<<'EOF'
<!doctypehtml>
<htmllang="en">
<head>
  <metacharset="utf-8">
  <metaname="viewport"content="width=device-width,initial-scale=1">
  <title>CloudSecurityLab</title>
</head>
<body>
  <h1>WelcometomyEC2WebApplication</h1>
  <p>Unit3-CloudSecurityAssignment</p>
  <p>HostedusingApacheonAmazonEC2.</p>
</body>
</html>EO
F
sudo chown root:root/var/www/html/index.html
sudo chmod 644 /var/www/html/index.html
```

Step 5 — Verify the webpage from theinstance

```
curl-Ihttp://localhostcurlh
ttp://localhost
```

Representativeoutput:

```
HTTP/1.1 200OK
Date:<server-generateddate>Server:
Apache
Content-Type: text/html; charset=UTF-8

<h1>Welcome to my EC2 WebApplication</h1>
<p>Unit 3 - Cloud SecurityAssignment</p>
<p>Hosted using Apache on AmazonEC2.</p>
```

Step 6 — Verify in abrowser

Fromadevicethatcanreachtheinstance,open**http://<EC2-PUBLIC-IP>**.Thebrowsershoulddisplaythe heading “WelcometomyEC2WebApplication”.Ifitdoesnot,verifytheinstancehasapublicIP,theHTTP rule permits your test source, and Apache is active.

5. Expected Results andValidation

- TheEC2instanceisintheRunningstateanditsstatuscheckspass.
- SSHaccesssucceedsusingthecorrectprivatekeyandrestrictedsourceIP.
- TheApacheserviceisenabledandactive.
- curlreturnsHTTP200OKanddisplaystheHTMLcontent.
- Thepublicbrowserrequestdisplaysthedeployedwebpagewhenthenetworkandsecurityrulespermitit.

6. Troubleshooting

SSHtimeout	Checkinstancestate,publicIP,routetable/internetgateway,andTCP22 source IP rule.
Permission denied(publickey)	Checktheloginuser(ec2-userforAmazonLinux),keypair,andprivate-keypath/permissions.
Website notloading	CheckTCP80inboundrule,Apachestatus,instancepublicIPv4,andsubnetrouting.
HTTP403/404	Checkthedocumentrootandfilepath;reviewApachelogswithsudo journalctl -u httpd.

curl command output showing HTMLresponse


```
gfg0622@Anshu-ka-lappi:~$ curl https://example.com
<!doctype html>
<html>
<head>
    <title>Example Domain</title>

    <meta charset="utf-8" />
    <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <style type="text/css">
    body {
        background-color: #f0f0f2;
        margin: 0;
        padding: 0;
        font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe UI"
```

7. Clean-up

Afterevaluation,stoporterminatetheinstanceasinstructedbythefaculty.Terminatingdeletestheinstance and can remove associated ephemeral storage; review attached volumes and other billable resources. Do not delete a resource needed for grading before confirming submission requirements.

8. Conclusion

ThispracticaldemonstrateshowtohostabasicHTMLapplicationonanEC2LinuxinstanceusingApache.It also demonstrates the importance of security-group rules, key-based SSH authentication, service verification, and testing the website through both curl and a web browser.

9. References

- AmazonEC2UserGuide:https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EC2_GetStarted.html
- Connect to a Linux instance using SSH:https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/connect-linux-inst-ssh.html
- Securitygrouprules:https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/security-group-rules.html
- AmazonLinux2023UserGuide:https://docs.aws.amazon.com/linux/al2023/ug/
- ApacheHTTPServerdocumentation:https://httpd.apache.org/docs/